

Basic Principles of Biology

Introduction to Basic Principles of Biology

What is Biology?

Biology as a Subject

Biology as a subject is a study that unifies much of natural science, It's a study of a point of convergence between the tools and information of all natural sciences. Biological system are the most complex chemical systems on earth, with it's capabilities and principles constrained by chemistry and physics.

System Biology

System Biology is an approach to model the dynamic behavior of biological systems based on it's study of the interaction between the system parts.

So What is Biology? *THE SCIENCE OF LIFE*

Biology is the study of all living things or *THE SCIENCE OF LIFE*

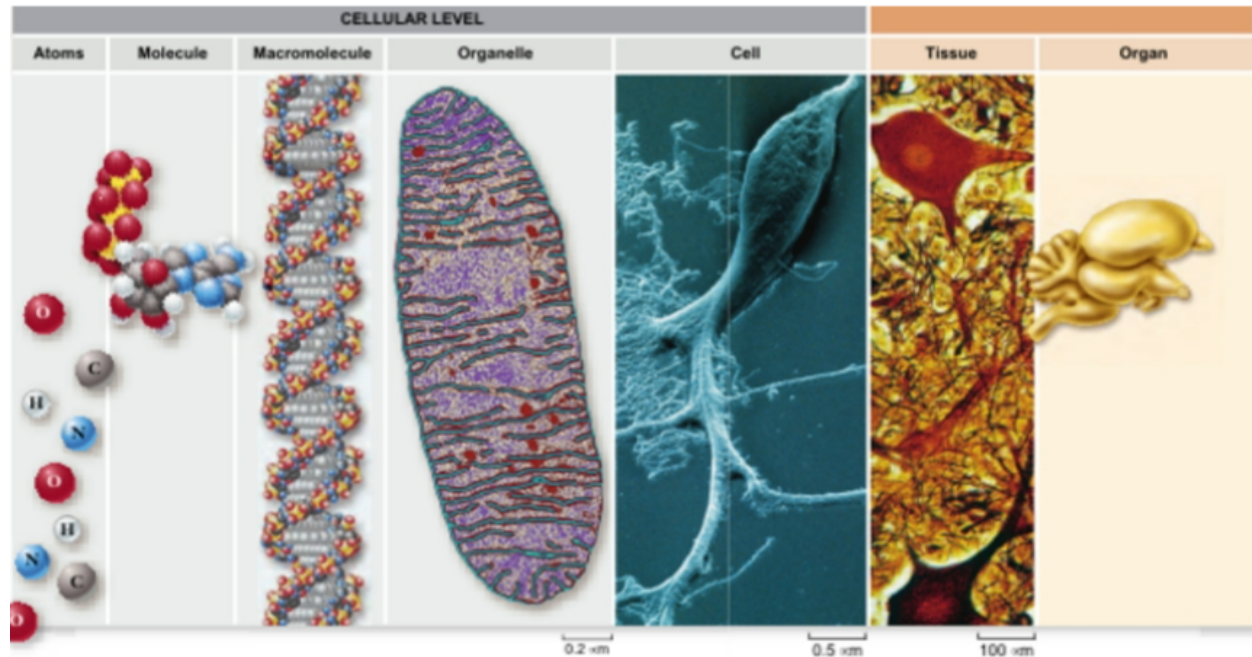
7 Character Shared by Living System

1. Cellular Organism : All organism is comprised of 1 or more cells that's bounded by membrane to carry out various task.
2. Ordered Complexity : All living things are complex and highly ordered.
3. Sensitivity : All organism respond to a stimuli, like a plant that search for sunlight.
4. Growth, Development, and Reproduction : All organism is capable of growing and reproducing, with it passing down hereditary molecules to ensure the offspring is of the same species.
5. Energy Utilization : All organism use energy to do some kind of work.
6. *Homeostasis* : All organism keep some kind of internal condition that's relatively constant and different from the environment.
7. Evolutionary Adaptation : All organism interact with the environment, both living and non living in ways that influence it's survival, and as consequence of it, it'll evolve to adapt to the environment.

Living System Show Hierarchical Organization

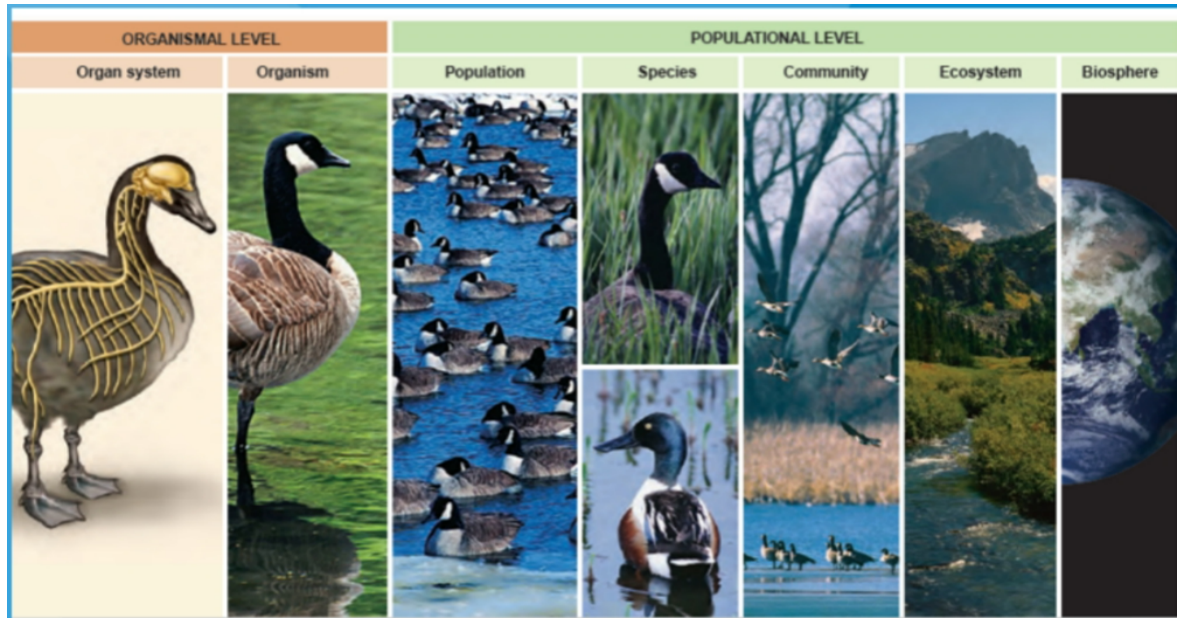
1. Cellular Level : At this level atoms are joined together creating **molecules**. Complex molecules are assembled into **organelles** within a **cell**. With some organism being

single cells, like bacteria and some multicell, like human and goat.



2. Organismal Level : Cells in a complex multicellular organism have 3 different levels of organization :
 1. Tissues : Groups of similar cells that act as a foundational units.
 2. Organ : Group of several different tissues that act as a structural and foundational units.
 3. Organ Systems : Group of several organs that created a system to ensure it's viability.
3. Populational Level : Hierarchies of a that categorizes group of individual organism. With it's hierarchies being :
 1. Population : a group of individual organism of the same species and living at the same place.
 2. Species : The compilation of all population of a particular kind of organism.
 3. Biological Community : All population of all different species living in the same space.
 4. Ecosystem : The biological system and physical habitat of it, creates a ecological system.

5. Biosphere : The planet that houses many kinds of ecosystem.



Unifying Themes in Biology

1. Cell Theory : The Organization of Living Systems with cells being the foundational units of life.
2. Molecular Basis of inheritance explains the continuity of life founded in DNA as a hereditary information.
3. The relationship between structure and function underlies living system.
4. Evolutionary conservation explains the unity in living systems (**ALL LIFE FROM A SINGLE SOURCE/COMMON ANCESTORS**)
5. Cells are information processing systems
6. Living systems exist in a nonequilibrium state -> Organism need constant support of energy to function.

Natural Selection

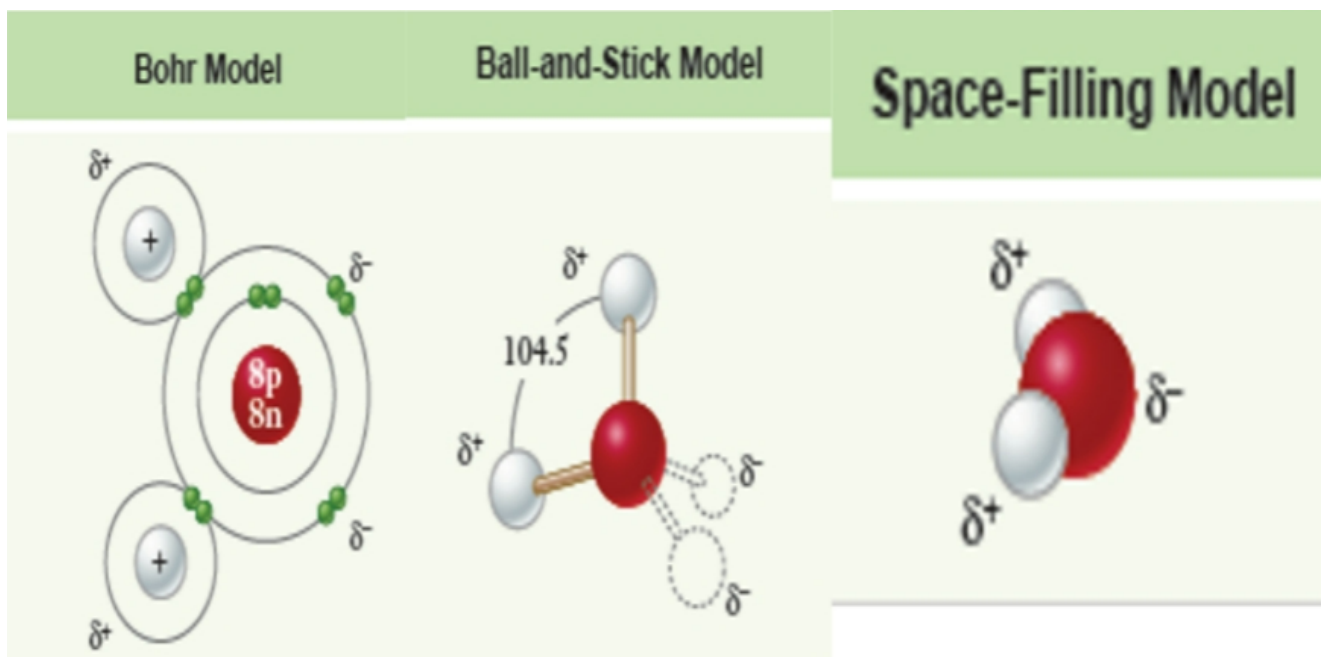
Natural Selection is the mechanism of evolutionary adaptation to selects the increase of a certain traits among multiple natural occurring traits.

The Building Blocks of Life

Water and Life

Water : Vital Compound

Water is a molecule of a partially positive H in 1 molecule and a partially negative O in another molecule, denoted as H₂O because it has 2 Hydrogen Atoms and 1 Oxygen Atoms.



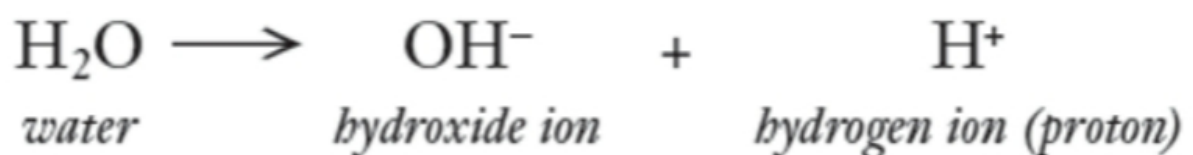
Characteristic of Water Molecules

Water has some unique characteristic :

1. Water Molecules are Cohesive : Water molecules tends to adhere to 1 another, this is a reason for **surface tension**.
2. Water Molecules are Adhesive : Water molecules tends to stick to other polar molecules.

Properties of Water

1. Water's High Specific Heat Helps in Maintaining Temperature
2. Water's High Heat of Vaporization helps in Cooling
3. Solid water is less dense that liquid water
4. The Solvent properties of water help move ions and polar molecules :
 1. Polar Molecules are attracted to water (**Hydrophilic**)
 2. Molecules that's non polar are repelled by water (**Hydrophobic**)
 3. The tendency of non polar aggregation in water is **Hydrophobic Exclusion** which can effect the structure of DNA, proteins, and biological membranes.
5. Water can forms ions :



Acid and Bases

pH Levels

- pH Scale measures the concentration of H^+ . With pure water having pH levels of 7 and 1 difference of pH measures 10-fold change in H^+ .
- Buffers help stabilize pH, Carbon dioxide and water react to create carbonic acid that resist changes in pH by absorbing and releasing H^+ . In human blood the buffer is carbonic acid/bicarbonate pair.

Acids

Any substance that dissociates in water to increase H^+ and lower the pH. The stronger an acid the more it ionizes water. for example Hydrochloric Acid (HCl) in the stomach completely ionizes water.

Bases

Any substance that combines with H^+ when dissolved in water and increase the pH, thus lowers the H^+ . Basic alkaline have a pH more then 7 with some like Sodium Hydroxide (NaOH), have pH of 12 and more.

Carbon : The Framework of Biological Molecules

Carbon and Hydrocarbons

- **Carbon** as the backbone of all biological molecules, can form 4 covalent bonds and make long chains.
- **Hydrocarbons** consist of carbon and hydrogen, with it's bond storing considerable amounts of energy.

Isomers

- Molecules with the same molecular or empirical formula
 - Structural Isomers
 - Stereoisomers
- Enantiomers
 - Mirror Image Molecules
 - Chiral
 - D-Sugars and L-Amino Acids

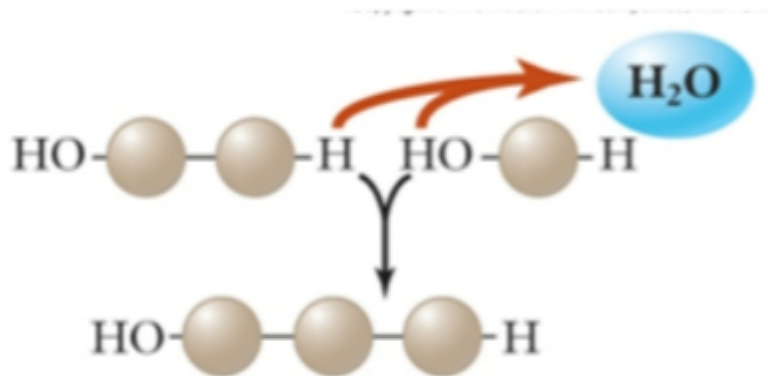
Macromolecules

- Polymer : Built by linking monomers
- Monomer : Small, similar chemical subunits
- Important macromolecules and their usage
 - Starch/glycogen is used in energy storage
 - DNA is used to encode genes (chromosomes)

- RNA is needed for gene expression (messenger RNA)
- Functional protein (amino acids) used to transport or as a catalyst (hemoglobin)
- Structural protein used to support the body (hair, silk)
- Fats (glycerol, 3 fatty acids) is used as energy storage (butter, corn oil, soap)
- Phospholipids is used as cell membranes (phosphatidylcholine)
- Steroids (four fused carbon rings) is used for membranes and hormones (Cholesterol, estrogen)

Dehydration Synthesis

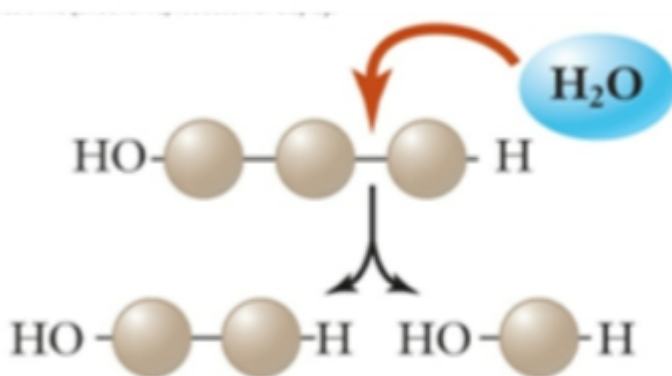
- Formation of large molecules by removal of water
- Monomers are joined to form polymers



a. Dehydration reaction

Hydrolysis

- Breakdown of large molecules by addition of water
- Polymers are broken down to monomers



b. Hydrolysis reaction

Carbohydrate

What is it?

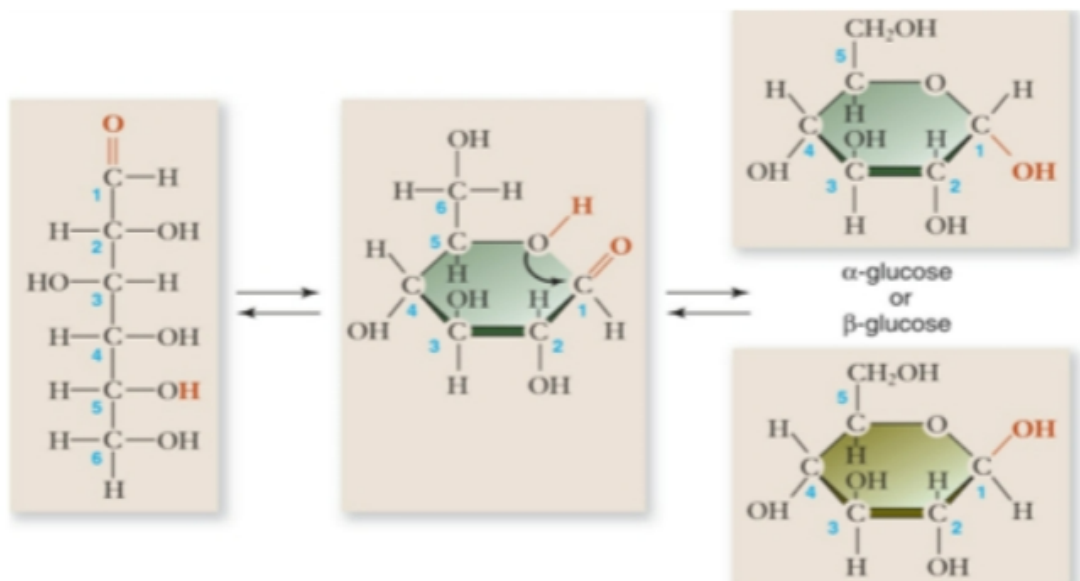
Carbohydrates are a loosely defined group of molecules that contain carbon, hydrogen, and oxygen with molar ratio of 1:2:1. Their empirical formula being CH_2O .

C-H Covalent bonds hold much energy, making carbohydrates great for energy storage. For example is sugar, starch, corn syrup, and more.

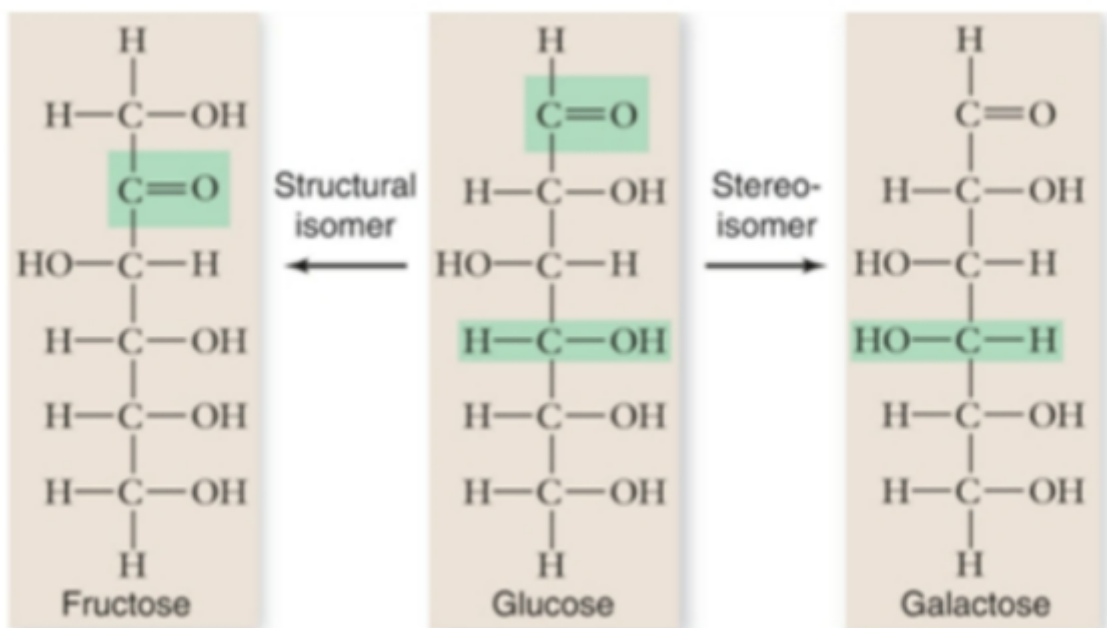
Type of Carbohydrate

Monosaccharides

- Simple Carbohydrates
- 6 Carbon sugars play important roles
- Glucose $\text{C}_6\text{H}_{12}\text{O}_6$.

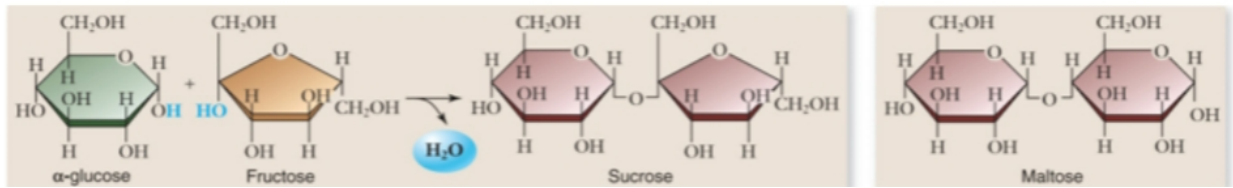


- Fructose is a structural isomer of glucose.
- Galactose is a stereoisomer of glucose.
- Enzymes that act on different sugars can distinguish structural and stereoisomers of this basic six-carbon skeleton



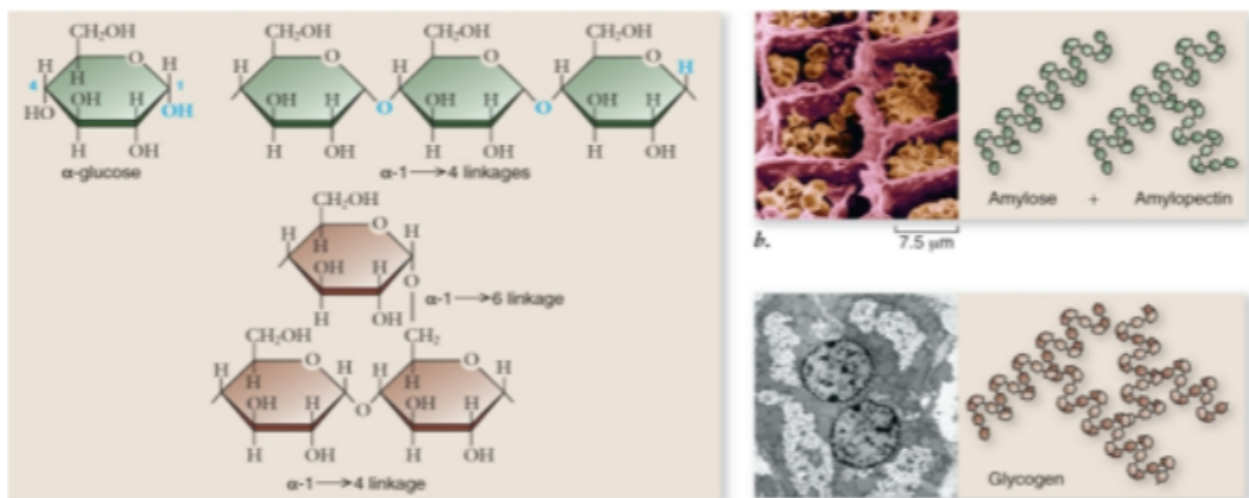
Disaccharides

- 2 monosaccharides linked together by dehydration synthesis.
- Used for sugar transport or energy storage.
- Examples : Sucrose, Lactose, Maltose

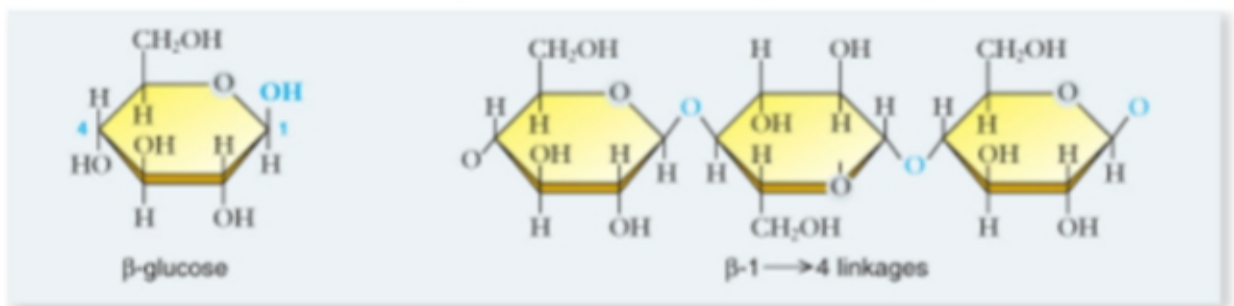


Polysaccharides

- Long chains of monosaccharides that's linked through dehydration synthesis.
- Energy storage, for example : plant use starch



- Structural Support
 - Plants use cellulose
 - arthropods and fungi use chitin



Nucleic Acids

- DNA and RNA are polymers composed of nucleotide monomers.
- Cells use nucleic acids for information storage and transfer.
- **Nucleic Acids are nucleotide polymers** : With it containing 4 nucleotide bases, in DNA there's adenine, guanine, cytosine, and thymine. in RNA thymine is replaced by uracil.

- Nitrogenous Bases include :
 - Purines : adenine & guanine
 - Pyrimidines : thymine, cytosine, uracil
- Nucleotides are connected by phosphodiester bonds.

DNA

Encodes information for amino acid sequence of proteins.

Have a Double Helix structure -> 2 Polynucleotide connected by hydrogen bonds.

- Base Pairing rules :
- A and T (U in RNA)
- C and G

RNA

Similar to DNA excepts :

- Contains ribose instead of deoxyribose
 - Contains uracil instead of thymine
- Single polynucleotide strand and use information in DNA to specify the sequence of amino acids in proteins.

Protein

What is Protein?

Protein are monomers that's composed of 1 or more long unbranched chains of polypeptide.

Amino Acid are monomers with a structure of

- Central carbon atom
- Amino group
- Carboxyl Group
- Single Hydrogen
- Variable R Group

Several examples of protein :

- Enzymes catalyst: proteases (breaks down protein)
- Defense : immunoglobulins/antibodies
- Transport : hemoglobin
- Support : Keratin and Fibrin
- Motion : muscle
- Regulation : Albumin, insulin
- Storage : Ferritin, casein

4 Level of Protein Structure

1. Primary Structure : Sequence of amino acids
2. Secondary Structure : interaction of groups in the peptide backbone :
 - a Helix
 - b HelixMotifs are common elements of secondary structures, seen in many polypeptides. It is useful in determining the function of unknown proteins.
3. Tertiary Structure : Final folded shape of globular protein :
 - Stabilized by a number of forces
 - Final levels for protein consisting of single polypeptide chains
4. Quaternary Structure : Arrangement of individual chains in a protein with 2 or more polypeptide chains.

Additional Structural Characteristics :

- Domains : Functional Units within a larger structure with most proteins made up from multiple domains that perform different parts of the protein function.
- Chaperones : Helps protein fold correctly and deficiencies in it, implicate a **Cystic fibrosis** which is a genetic disorder.
- Denaturation : Protein loses structure and formation due to environmental conditions.

Lipids

Loosely defined group of molecules that are insoluble in water, because of its high proportion of C-H that causes it to be hydrophobic. examples are fats, oil, waxes, and more.

Fats

Type of Fats :

- Triglycerides which's composed of 1 glycerol and 3 fatty acids.
- Fatty Acids :
 - Is not identical
 - chain length varies

Phospholipids

Composed of Glycerol, 2 fatty acids with non polar tails, and a phosphate group with polar head. Phospholipids form all biological membranes.

Watson-Crick base pairing